



Operating, material and service plan

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Record origin: User-authorized fictional case implementation, derived from controlled handoff and explicitly labeled engineering assumptions. Official sources establish external context only; they do not verify fictional assets, transactions, operating events or professional approvals. References observed September 6, 2026 UTC / September 5 PDT unless the pinned source manifest records an earlier acquisition.

BS&T's 2025 base is 9,000 loaded revenue cars plus an estimated 8,100 empty movements over 260 service days. A loaded revenue car and an empty return are different measures; they must not both be counted as revenue carloads. The 40-mile estate includes the main and both branches. Rail capacity is conditional on train formation, interchange paths, yard occupancy, equipment availability and rested qualified crews.

The normal road assignment pairs the two owned GP40-2 locomotives. The GP38-2 is a protected spare and occasional yard unit; the SW1200 is stored pending repairs. No acquisition-day locomotive lease is assumed. Nameplate horsepower and the declared tractive-effort allowances are equipment-model assumptions, not verified individual locomotive acceptance tests. Mechanical records must preserve that distinction.

A first 18-loaded-car calculation failed the declared ruling-grade traction test. At a conservative 143 short tons gross per loaded car, the corrected ceiling is **16 loaded cars** with the paired road locomotives. The model includes 260 short tons for the two locomotives, a 1.8% formation grade and 5 lb per ton rolling-resistance allowance. Required effort is 104,468 lb against the model's 109,400 lb pair allowance. That narrow reserve means wet rail, wind, curvature or degraded equipment requires tonnage reduction; the timetable must not promise 16 cars under every condition. The spare substitution likewise requires a fresh tonnage calculation.

Three ordinary round trips supply 48 loaded slots per day against about 34.62 required loaded cars. The screening round-trip allowance is 3.33485 running hours at a 20 mph average plus 1.2 hours switching, giving about 13.60 daily mainline train hours. A further 1.5 hours covers one daily round trip on each branch at 15 mph and local switching, giving **15.10 total train hours**. Separate eight-hour crew turns cover the service with relief and statutory rest. The 22 train-and-engine employees also cover yard assignments, leave, qualification, extra board and relief; there is no assumption that one crew works the whole operating day.

The 5,000–7,500 additional annual-car scenario needs four daily round trips, 64 loaded slots, about 19.64 train hours including branch service, three staggered qualified crews, agreed interchange capacity, completed track/yard capital and the protected spare. At the high case, 16,500 annual loaded cars require 63.46 per day, so the model has little slot reserve. The high figure is therefore conditional network capacity, not committed Red Wash demand or automatically available service during a locomotive failure. Lower daily loading or additional service days would be needed for resilience at that upper bound.

Red Wash's normalized inbound basis is **205 ordinary billable cars, ten project cars and ten contingent unbilled slots**. That equals the 225-car planning allowance; 300 is capacity. Contingency is neither earned revenue nor an invented delivered commodity. The annual physical book is 68 acid, 80 binder, 17 lime, 28 steel, 12 MRO and ten project cars. Finished uranium product is excluded.

The physical payload model is 100 short tons for acid/binder/lime, 75 for steel, and 40 for MRO/project cars. Ordinary drayage uses 25-ton acid/dry-bulk/steel road payloads and smaller mixed/project consignments where appropriate. It therefore requires multiple truck trips per railcar, rather than treating a tank car as one road load. The exact

normalized annual tonnage and trips are generated in `interface_economics.json`.

At 225,000 ore tons and 330 processing days, 6,800 tons of 98% acid imply 20.606 as-received tons per processing day and about 29.62 kg pure acid per metric ton of ore. This is a synthetic metallurgical input; carbonate-rich feed can increase acid use nonlinearly and remains a diligence risk. The minimum acid inventory is 210 tons, with a 290-ton reorder point and 340-ton working inventory allowance. Two 30,000-gallon nominal tanks provide additional physical ullage within the declared density/usable-volume assumptions. The inventory policy protects roughly 10–14 processing days while admitting an incoming 100-ton rail batch through four road trips.

Binder consumption is 8,000 tons against 185,000 tons of dry backfill solids, a modeled 4.32% ratio to those solids. Three 150-ton silos provide 450 tons capacity. Minimum inventory is 245 tons, reorder is 340 tons and consumption averages 24.242 tons per processing day. Actual paste recipes remain controlled by the mine's technical operator. The 600-ton/day nameplate does not by itself prove a future 225,000-ton operating year; the mine's throughput/capital bridge remains a separate explicitly modeled dependency.

Tuesday and Friday 08:00 Mountain time are the scheduled Taylor handling windows. The default SLA is **72 elapsed hours** from interchange receipt through empty release; steel receives 96. Processing budgets include transload, drayage, empty inspection and release. If the next scheduled window would miss the limit, dispatch uses the next available weekday exception. If that still misses, funded on-call handling starts within 24 elapsed hours. The runnable calendar uses UTC arithmetic for elapsed time, including both daylight-saving transitions. This is an explicit modeled operational commitment, not a clock exclusion or legal waiver.

The mine may permanently use outside carriers when ARU is unqualified or uncompetitive. Every receipt, spot, transfer, truck dispatch, mine delivery and empty release has a timestamp. Demurrage follows documented cause. No minimum volume guarantee or preference over external customers is granted. Direct uranium custody remains OPEN_GATED despite the July 7, 2026 start of qualified ordinary inbound service.

The limited 2026 ramp is 3, 6, 9, 12, 15 and 18 cars from July through December: 63 in the full-year forecast. At the September 5 cutoff, July and August are synthetic actual records, while the complete September month and the later months remain forecasts. The August 14 acid timestamp incident occurs after commencement and concerns ordinary inbound material, not unauthorized uranium movement.

The finance contract book assigns exactly **254,000 annual external terminal tons** across the Taylor and Rawlins sites. Adding 19,480 normalized mine tons uses 273,480 of 300,000 tons of combined capacity, leaving 26,520. Warehouse reservations average 11,750 of 15,000 slots. Generated contract-to-facility allocations retain the original customer and contract IDs. Rawlins has no BS&T rail branch. Rail receipts delivered to customer-controlled campus sidings are not automatically ARU terminal-handled tons.

The 15,000 external billed truck dispatches use an explicit short-haul mix averaging 25 loaded miles, an empty return, 40 mph and 0.75 hour handling: two hours per dispatch. They consume 30,000 annual hours; 788 normalized mine trips add 1,648. Twenty drivers supply 36,000 productive hours and 18 tractors provide 38,250 hours at the modeled duty/availability levels. The remaining four trucking employees are a manager, dispatcher and two mechanics. These assumptions support the inherited dispatch book; longer or unusually slow jobs require repricing or outside-carrier capacity, and must not be quietly substituted into the two-hour model.

The 15.10-hour ordinary timetable takes effect after the synthetic April 24, 2026 release of the MP 22–26 slow order. Before release, the four-mile 10 mph restriction adds 1.2 daily train hours across three round trips. Two paid crew turns of up to 8.5 hours cover the resulting 16.30 train hours, with legal rest and overtime preserved; the improved timetable is not backdated into the 2025 operating year.

Revenue-car supply primarily comes from connecting carriers, shippers and their lessors. The 12 owned maintenance cars and six leased bulkhead flats do not purport to carry all 9,000 annual revenue movements. A 2.5-service-day loaded dwell and 1.5-day empty dwell imply about 133 transient cars; adding the registered 18-car pool remains below 256 registered car spots. Wamsutter has three 20-car tracks, each 0.28 mile long, allowing the declared 60-foot reference cars and locomotive/clearance space. Actual train length remains a dispatch control.